```
    this.centroid[i] ← mean({point[i] for point in
points}                    #4
  this.radius ← max{distance(this.centroid, entry) +
entry.radius for entry in points} #5
```

#1 Method `updateBoundingEnvelope` is defined on `SsNode`. It updates the centroid and radius for current node.
#2 Get the centroids of the items inside the node's bounding envelope: for a leaf, those are the points held by the leaf, while for an internal node, the centroids of the node's children.
#3 Cycle through the `k` coordinates of the (`k`-dimensional) space.
#4 For each coordinate, computes the centroid's value as mean of the points' values for that coordinate; for instance, for the `x` axis, computes the mean of all `x` coordinates over all points/children in the node.
#5 The radius is the maximum distance between the node's centroid and its children's centroids. This distance includes the (Euclidean) distance between the two centroids, plus the radius of the children: we assume that points here have radius equal to `0`.

We need, of course, to compute the variance at each step of the for loop at line #4; perhaps this is the right time to remind you what variance is and how it is computed: given a set `S` of real values, we define its mean μ as the ratio between the sum of the values and their multiplicities: